

Topic: Microinstruction Sequencing and Execution

Q. Explain the concepts of Microinstruction Sequencing and Execution in a Microprogrammed Control Unit. What is Control Memory?

****Definition to Remember****

In a Microprogrammed Control Unit, the Microinstruction Sequencing and Execution are controlled by the Control Memory.

Microinstruction Sequencing & Execution

* It is a fast, Read-Only Memory (ROM) located inside the CPU, completely separate from main system memory.

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2. Microinstruction Execution

A microinstruction is basically a control word. The bits in this word directly translate to physical control lines in the CPU.

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3. Microinstruction Sequencing

Sequencing is the process of determining the address of the *next* microinstruction to fetch from Control Memory. The hardware unit responsible is the Sequencer.

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4. Workflow Loop

1. Map Machine Instruction Opcode -> Control Memory starting address.

2. Sequ

****Must-Write Points (for 10 marks)****

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Quick Recall

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